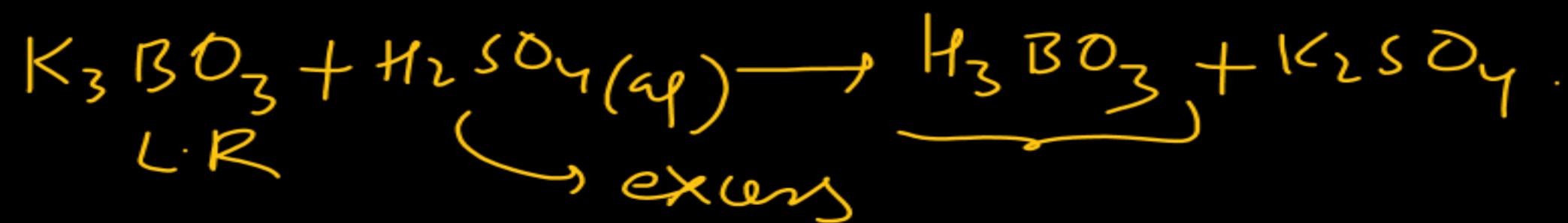
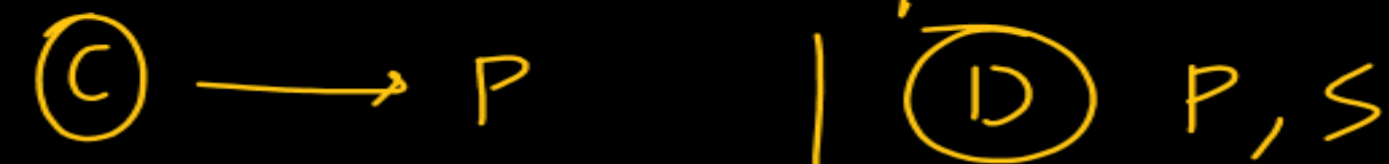
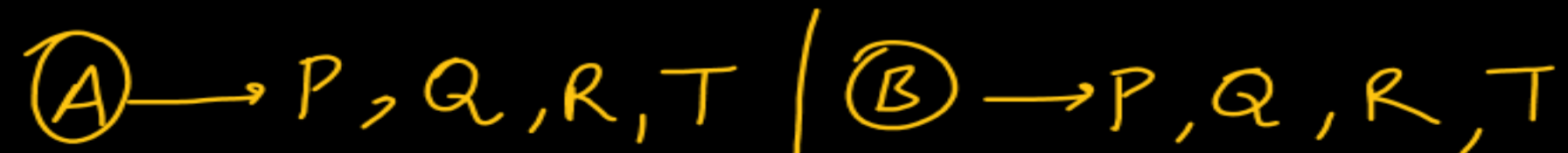
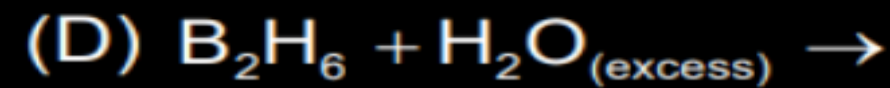
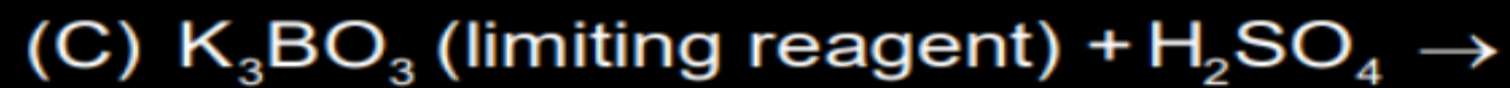
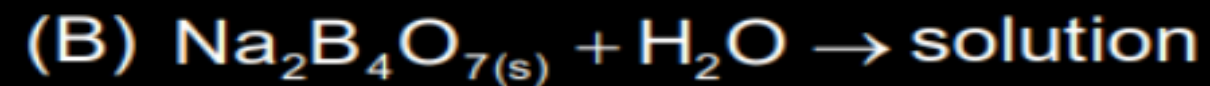
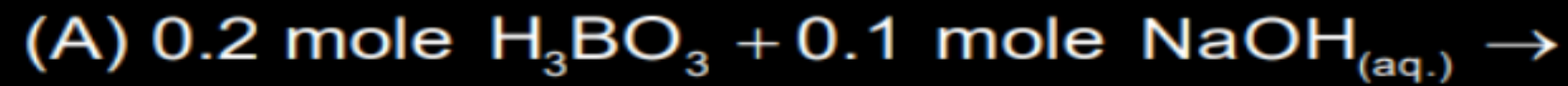


Match the matrix.

Column-I



Column-II

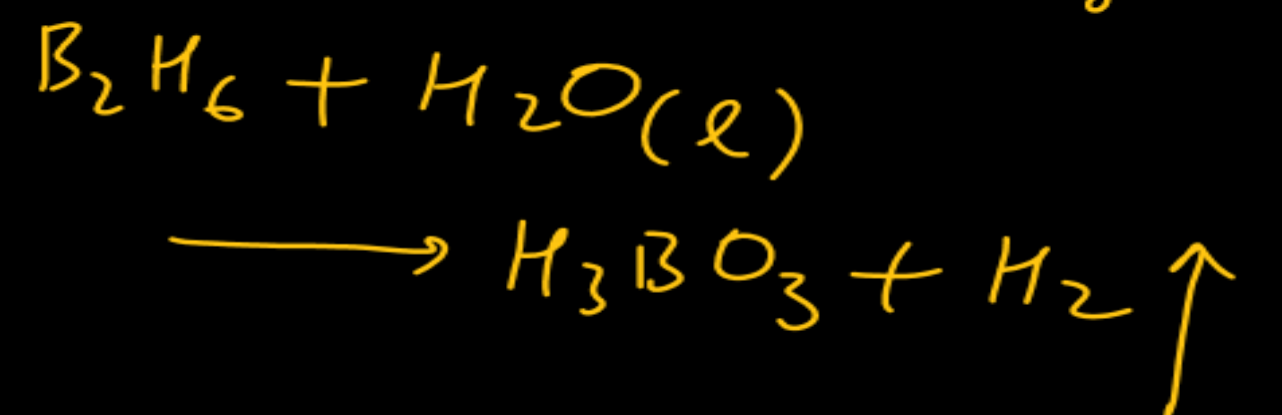
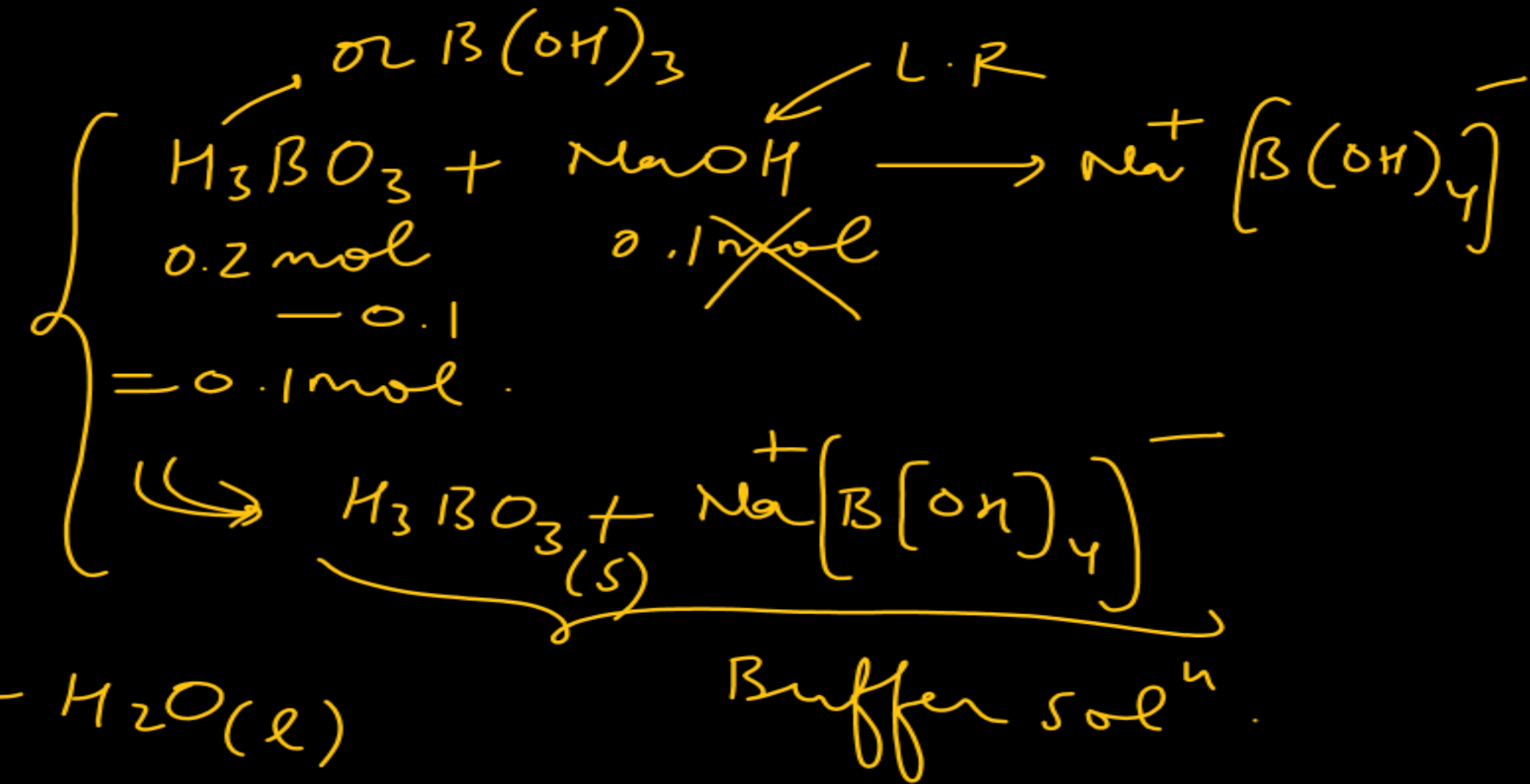
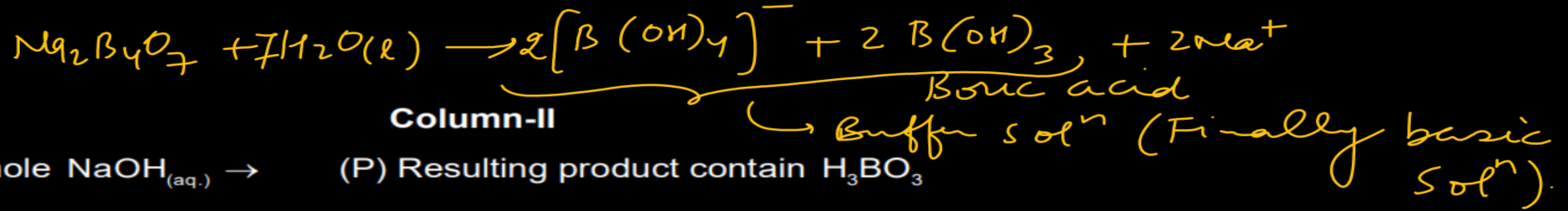
(P) Resulting product contain H_3BO_3

(Q) Alkaline solution

(R) Resulting solution is alkaline and contain $[B(OH)_4]^-$ in solution

(S) Volatile product

(T) Buffer solution



Match List-I with List-II.

List-I

[Property of a metal (M)]

(P) "M" produces H_2 with KOH solution

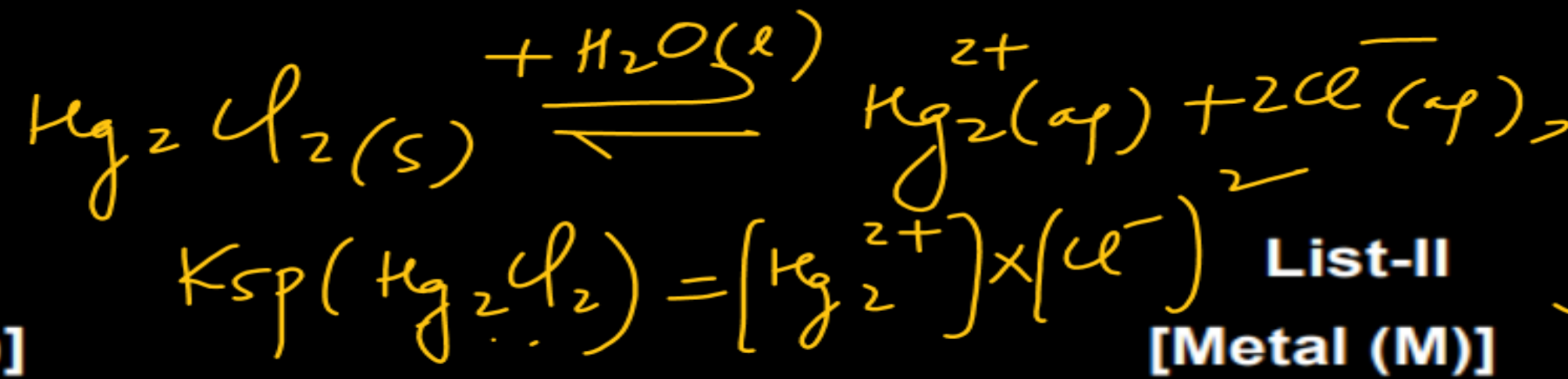
(Q) "M" produces M^{2+} ion when reacts with dil. HCl

(R) "M" does not dissolve in dil. HCl but dissolves in dil. HNO_3

(S) "M" does not dissolve in conc. HNO_3 , dil. HNO_3 or H_2SO_4

Select CORRECT code for your answer.

	P	Q	R	S
X (A)	1	3	2	4
X (B)	4	2	3	1
(C)	<u>2</u>	<u>4</u>	1	3
☒ (D)	<u>2</u>	<u>4</u>	<u>3</u>	<u>1</u>



List-II

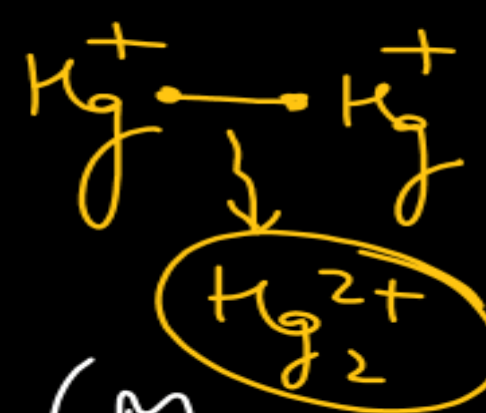
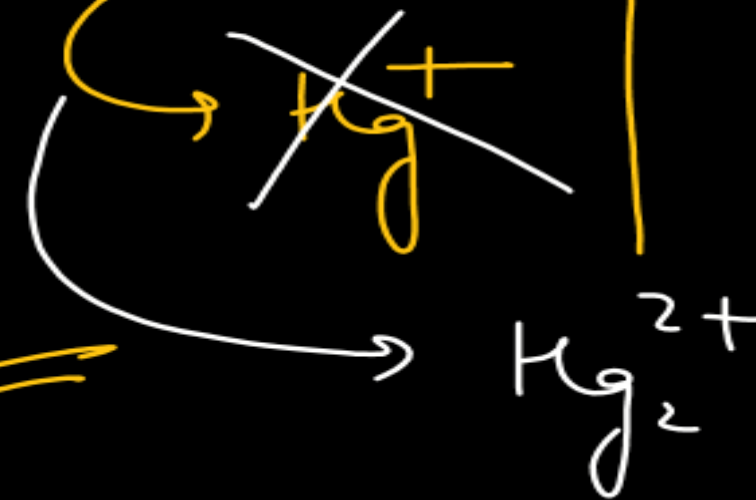
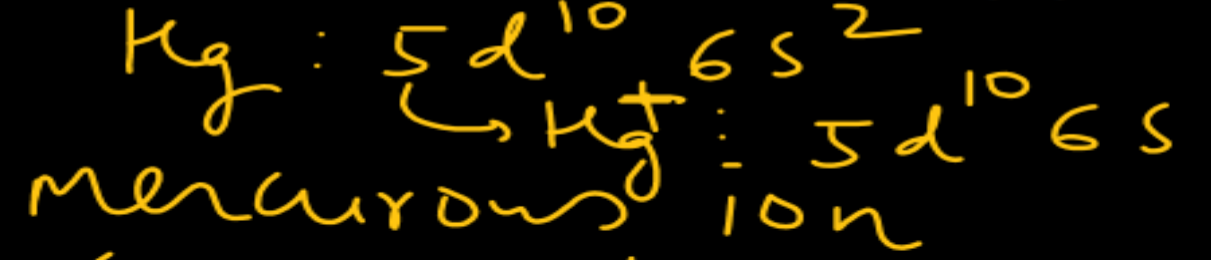
[Metal (M)]

(1) Au

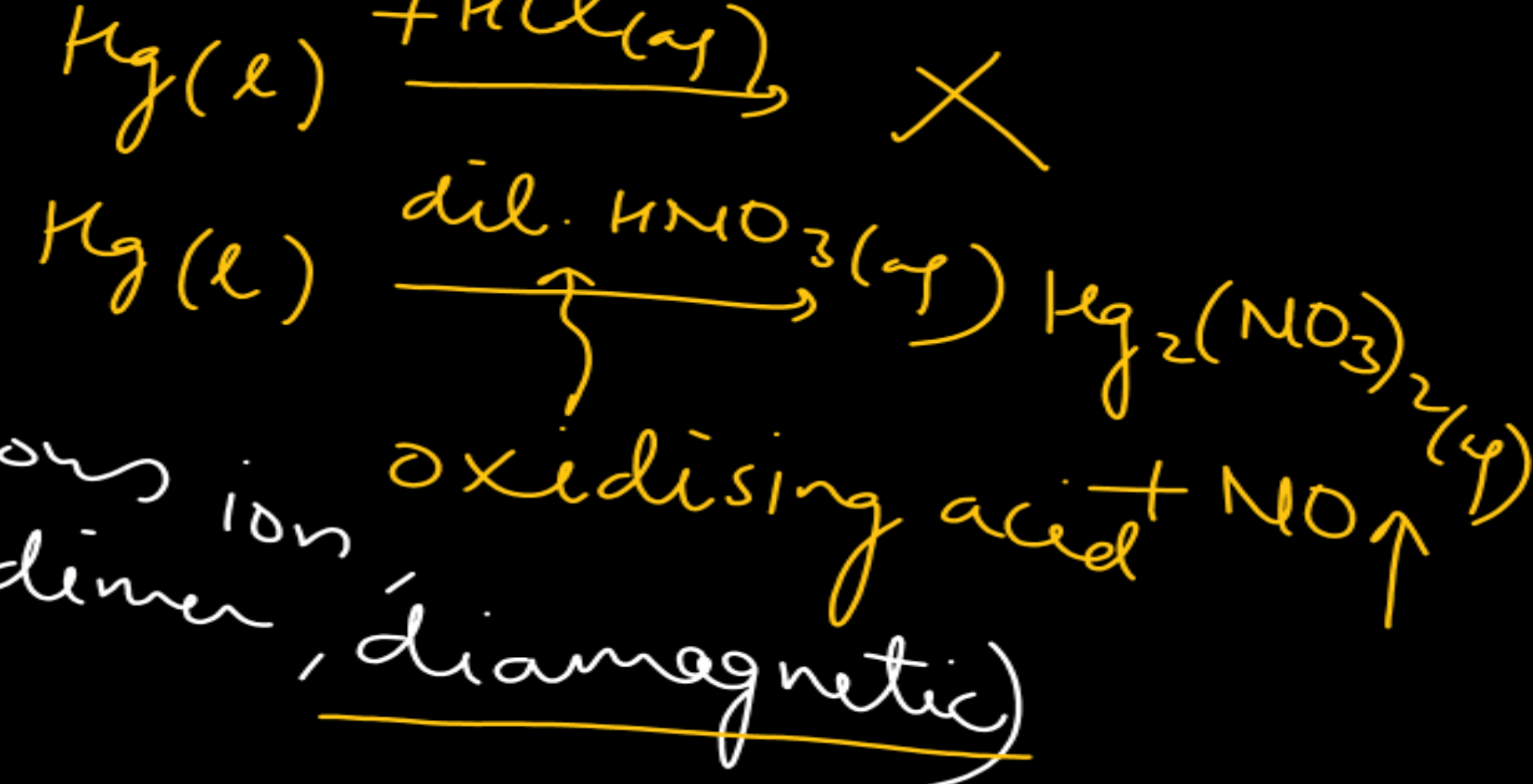
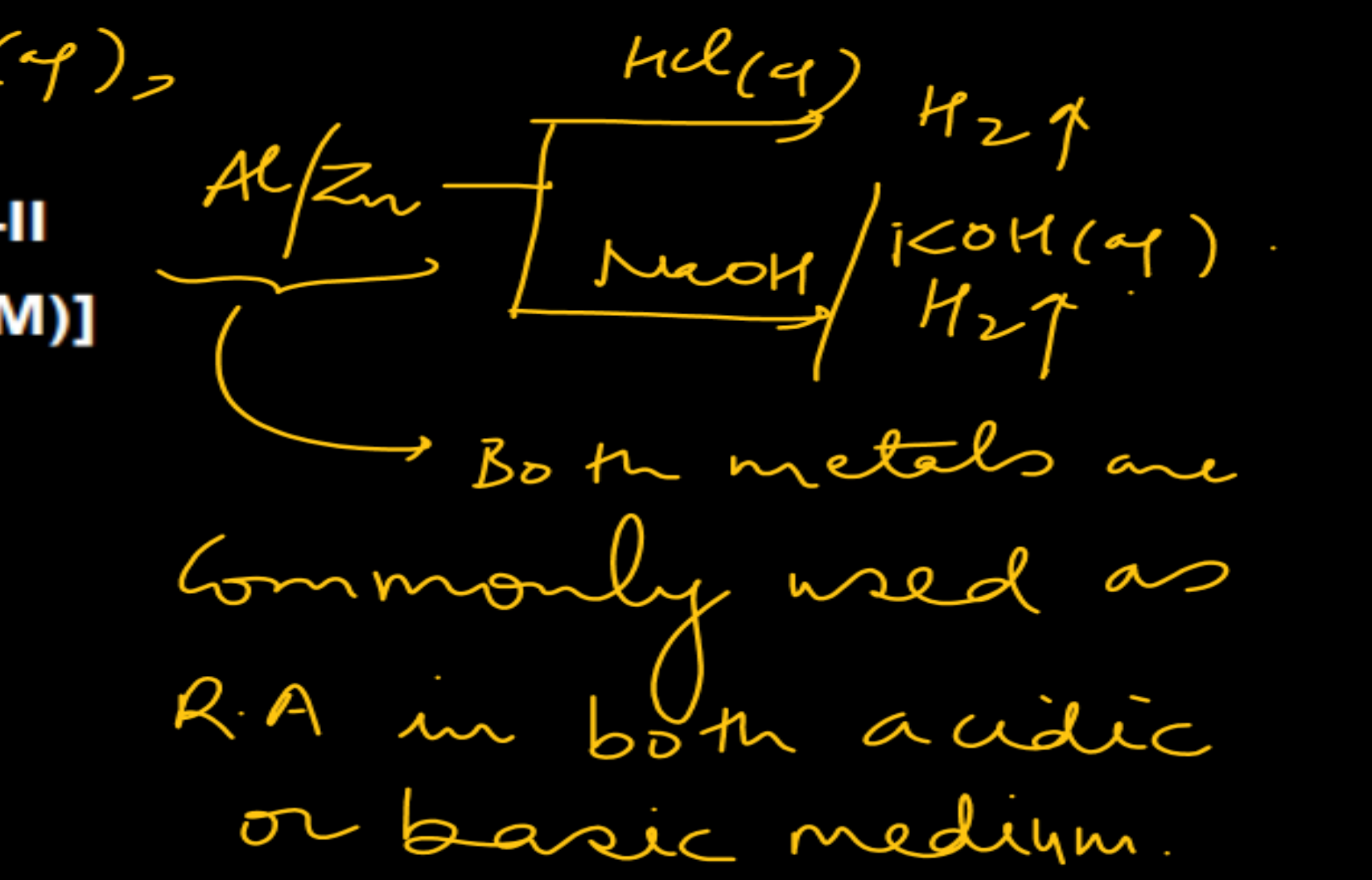
(2) Zn

(3) Hg

(4) Fe



(mercurous ion as a dimer, diamagnetic)



Hydrogen peroxide, when pure is an almost colourless (Very pale blue) liquid, less volatile than water.

H_2O_2 is miscible with water in all proportions and form hydrates $H_2O_2 \cdot H_2O$. Due to repulsive interaction of the O — H bonds with the lone pairs of electron on each O-atom H_2O_2 molecule show hindered rotation about O — O single bond.

⇒ Which property(ies) is/are higher for H_2O_2 as compared to H_2O (at 25°C)?

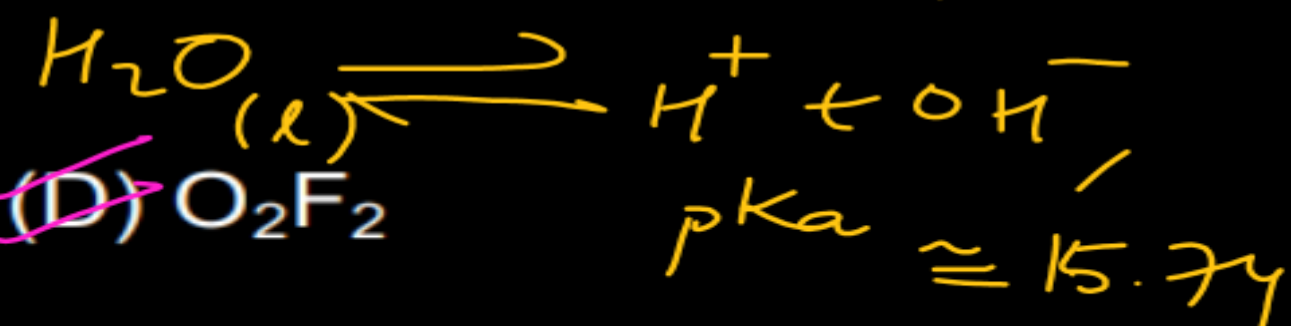
(A) Viscosity

(C) Vapour pressure

(B) Density

(D) Ionisation constant (K_a)

→ extent of H-bonding in H_2O_2 is more than that in H_2O .



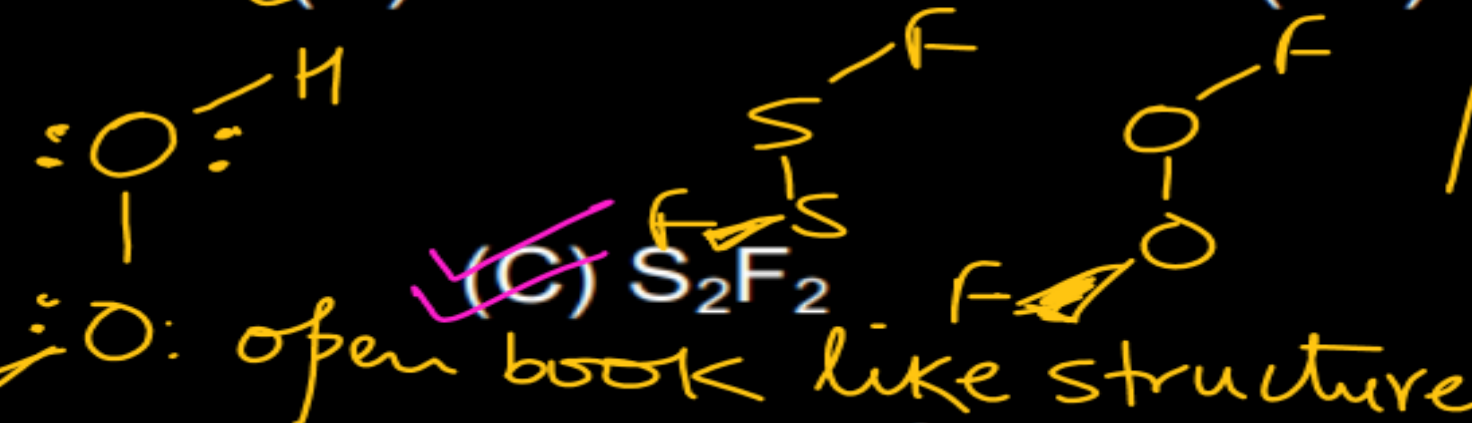
⇒ Which is/are isostructural with H_2O_2 ?

(A) C_2H_2

(B) N_2H_2

(C) S_2F_2

(D) O_2F_2



⇒ Which compound can produce H_2O_2 on hydrolysis in cold water?

(A) PbO_2 ✗

(B) MnO_2 ✗

(C) BaO_2

(D) Na_2O_2

(Barium peroxide)

$$(K_a(H_2O)) = 1.8 \times 10^{-16}$$

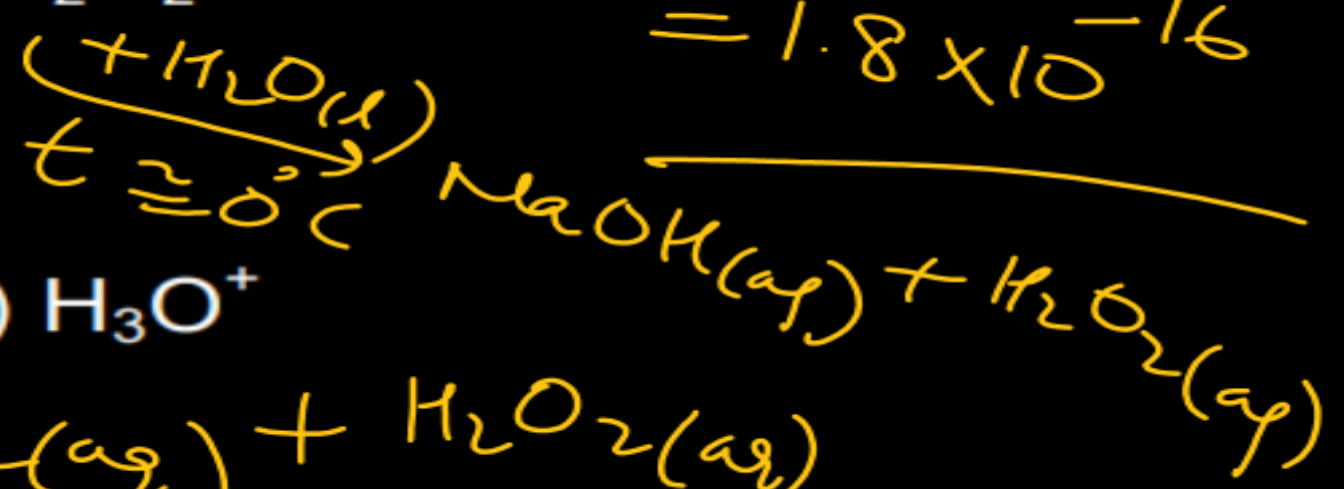
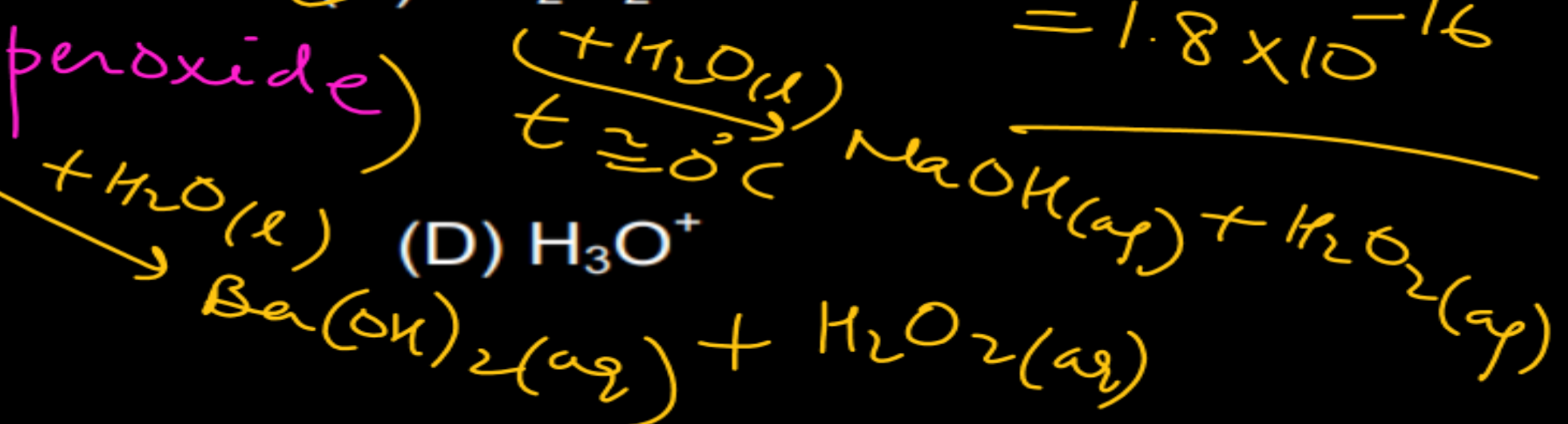
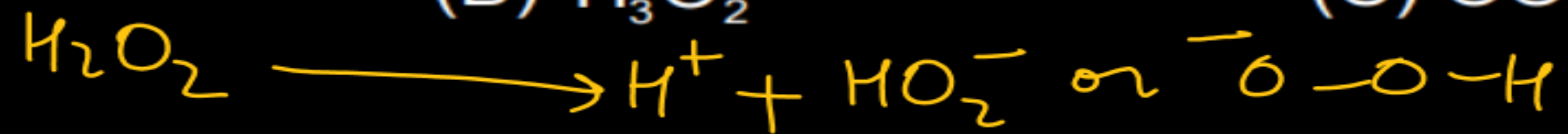
⇒ Deprotonation of hydrogen peroxide produces:

(A) $H_3O_2^+$

(B) $H_3O_2^-$

(C) OOH^-

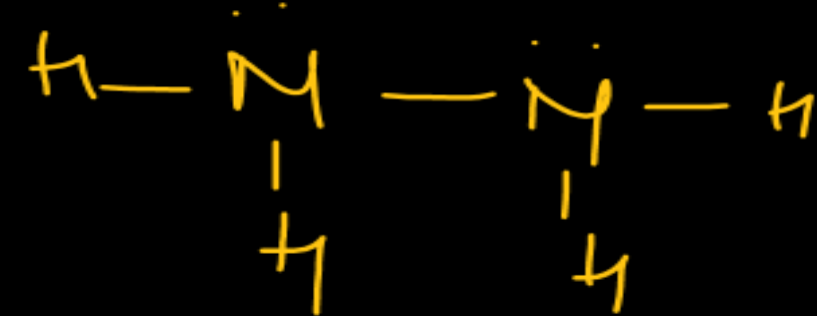
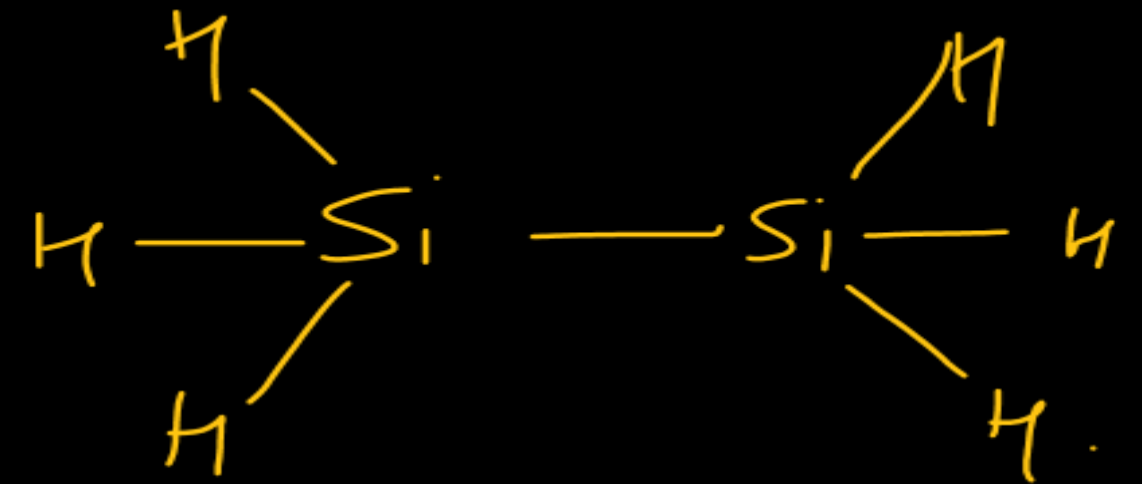
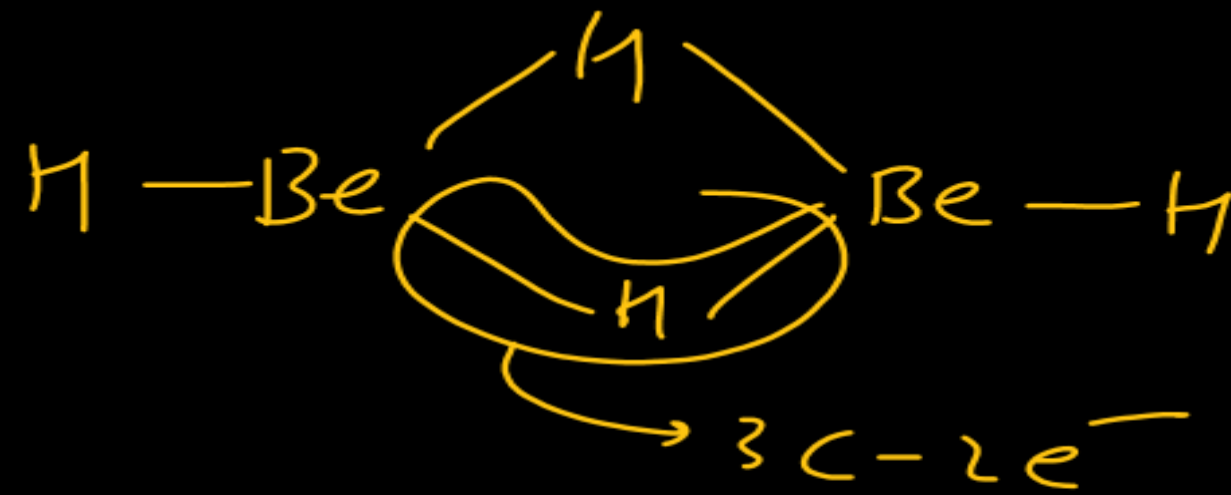
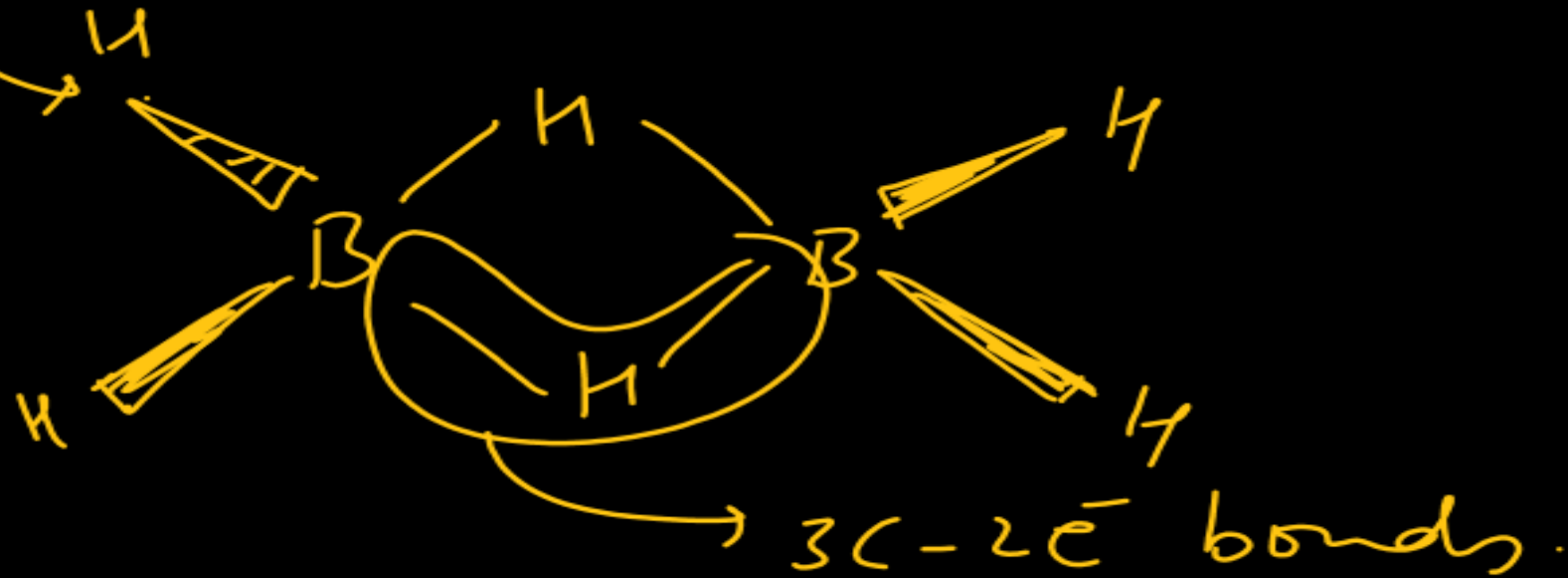
(D) H_3O^+

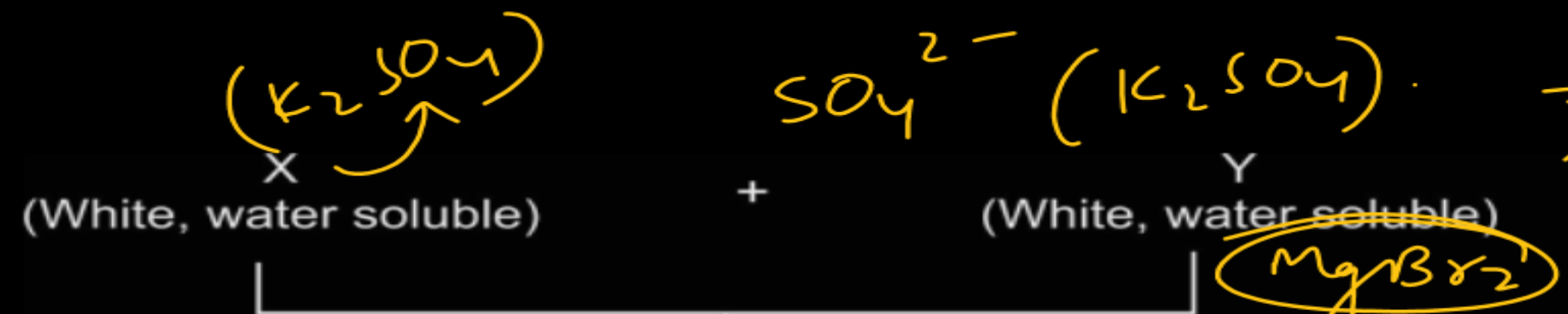


atoms ? ✓

✓ B₂H₆, Be₂H₄, Si₂H₆, N₂H₄, (BeH₂)_n, H₂O₂.

Ans: 3





$\Rightarrow$ X, X_1 and X_2 are respectively :

(A) Na_2SO_3 , NO_2 , NH_4ClO_4

(C) K_2SO_4 , Br_2 , $KClO_4$

(B) K_2SO_4 , NO_2 , NH_4ClO_4

(D) $(NH_4)_2SO_4$, Br_2 , $(NH_4)ClO_4$

H_2O
 Aqueous solution of mixture

Brown
 colour gas (X_1)

Cl_2
 Gas

White precipitate
 insoluble in dilute acid (Y_1)

$BaCl_2$

$NaOH$
 (Large excess)

ppt.
 (separated)
 (Y_2)

$Mg(OH)_2 \downarrow$
 white

Filtrate

$NaClO_4 (aq)$

Yellow precipitate
 (X_2)

$KClO_4 \downarrow$
 yellow ppt
 (X_2)

$Mg(OH)_2 + HCl(aq) \rightarrow MgCl_2(aq) + H_2O$

$Mg(OH)_2 + OH^- \xrightarrow{(excess)} No\ dissolution$

$\Rightarrow$ Y, Y_1 and Y_2 are respectively :

(A) $AlBr_3$, $BaSO_3$, $Al(OH)_3$

(C) $MgBr_2$, $BaSO_4$, $Mg(OH)_2$

(B) $Mg(NO_2)_2$, $BaSO_4$, $Mg(OH)_2$

(D) $AlBr_3$, $BaCO_3$, $Al(OH)_3$

$\Rightarrow$ Y_1 and Y_2 are soluble respectively in :

(A) dilute H_2SO_4 , $NaOH$

(C) dilute HCl , dilute H_2SO_4

(B) conc. H_2SO_4 , dilute HCl

(D) $NaOH$, dilute HCl

$BaSO_4 \downarrow \xrightarrow[conc.]{H_2SO_4} Ba(HSO_4)_2(aq)$
 (excess)

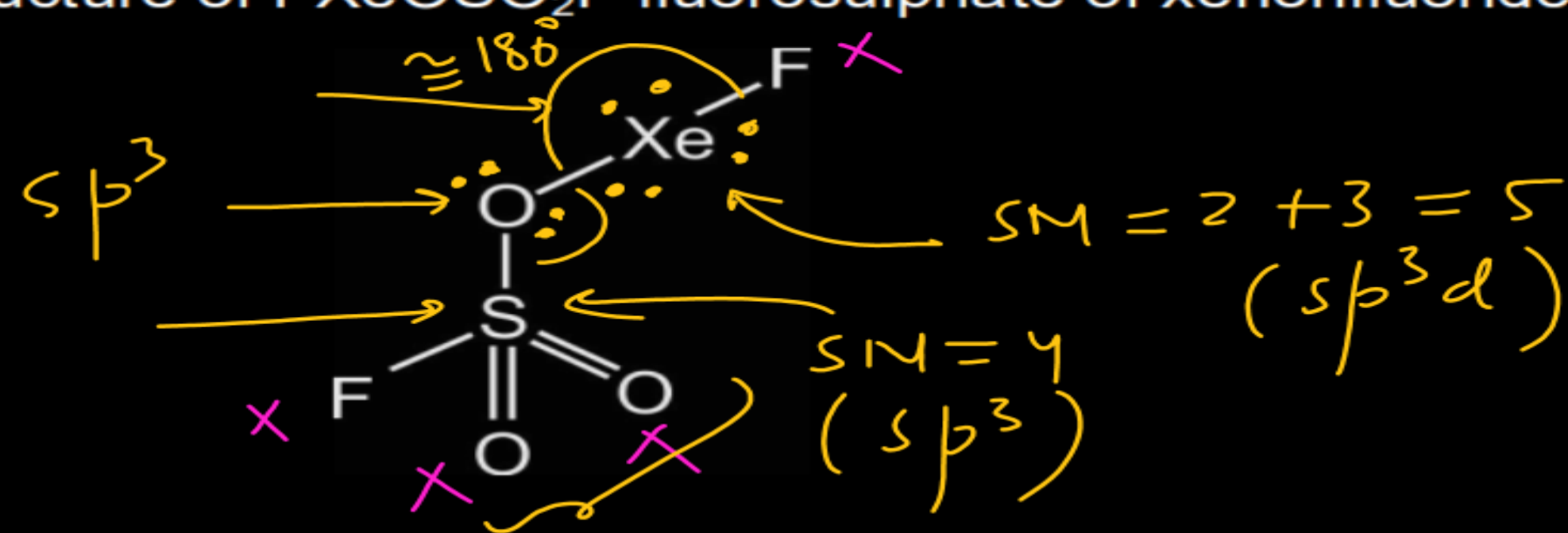
$NH_4^+ + NaOH$

$\rightarrow NH_3 \uparrow$

$Al(OH)_3 + excess\ of\ NaOH$

$\rightarrow NaAlO_2(aq)$

Molecular structure of FXeOSO_2F fluorosulphate of xenonfluoride is given below:



→ What is the hybridization state of Xe and sulphur respectively?

(A) sp, sp^3

(B) sp^3d, sp^3d

(C) sp, sp^3d

☒ (D) sp^3d, sp^3

→ Total number of lone pair(s) on non-terminal atoms in given molecule is:

☒ (A) five

(B) fifteen

(C) ten

(D) three

→ Select CORRECT statement(s) for given molecule. (b = bridging, t = terminal)

(I) $S - O_b$ bond length $>$ $S - O_t$ bond length ✓

(II) $F - Xe - O_b$ bond angle $>$ $Xe - O_b - S$ bond angle ✓

(III) Molecule has two $3p\pi - 2p\pi$ bonds ✗

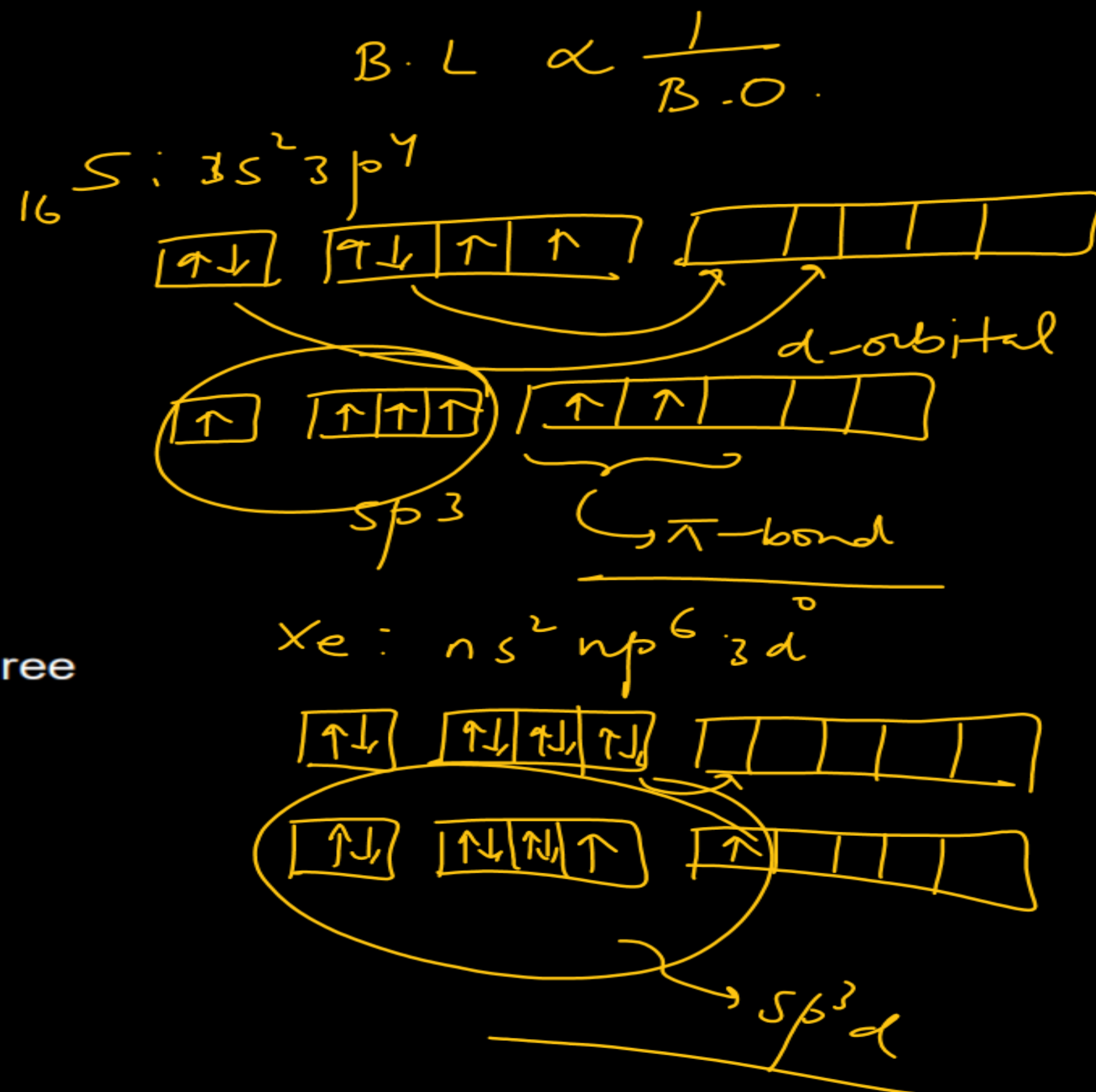
(IV) Total three d-orbitals are involved in bonding. ✓

☒ (A) I, II, IV

(B) I, II

(C) I, II, III

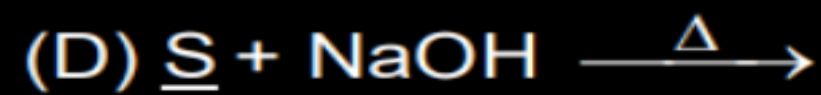
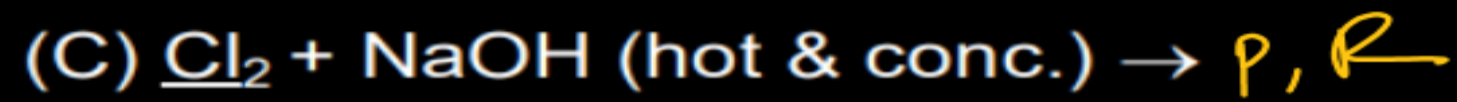
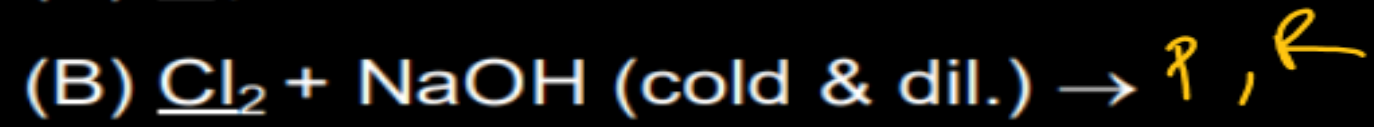
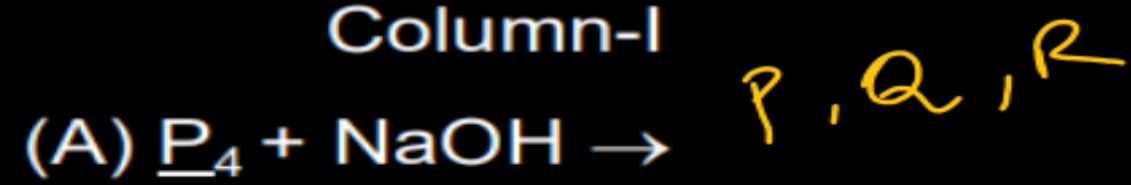
(D) I, II, III, IV



Match the matrix.

Column-I

Column-II



↪ P, R, T.

(P) Disproportion reaction

(Q) Gases product

(R) One of the products has lowest O.N. of underlined atom

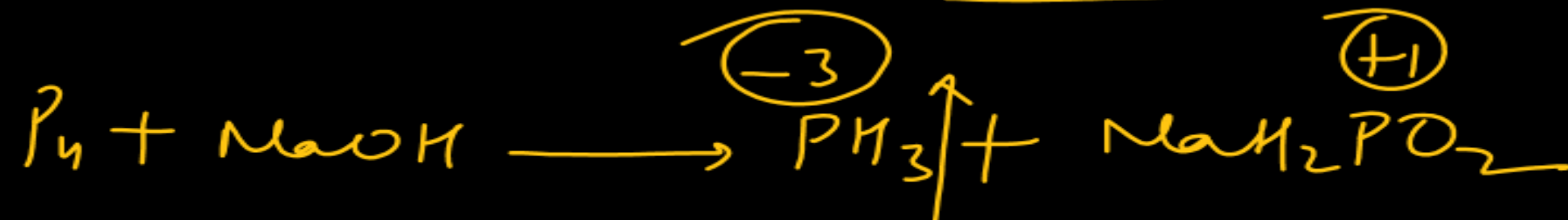
(S) One of the products has highest O.N. of underlined atom

(T) One of the products is used in fixing of Black and white

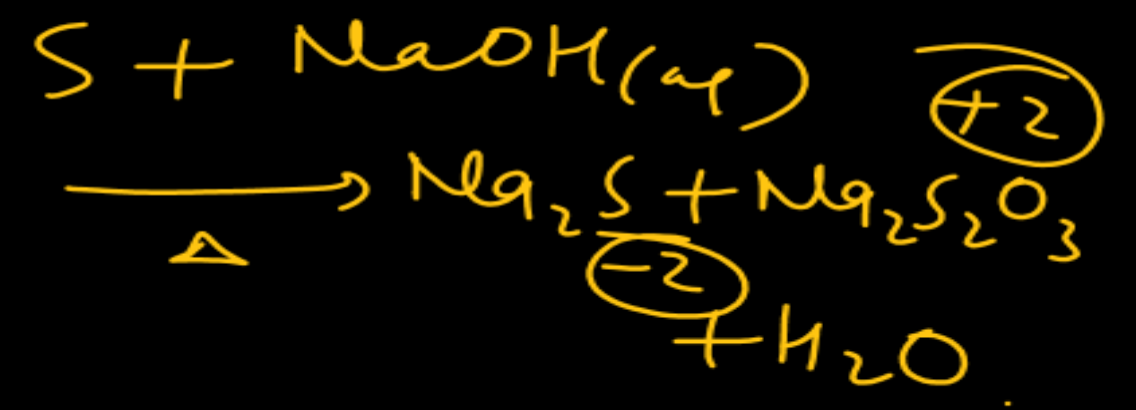
Photography

fixer : hypo solution : $Na_2S_2O_3(aq)$

sod. hypochlorite



↪ sod. chlorate



$\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ dissolve in water to form a complex species. If a solution prepared by dissolving 0.1 mole of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ in water is treated with excess silver nitrate solution, 0.2 mole of silver chloride is precipitated. Select CORRECT option for complex.

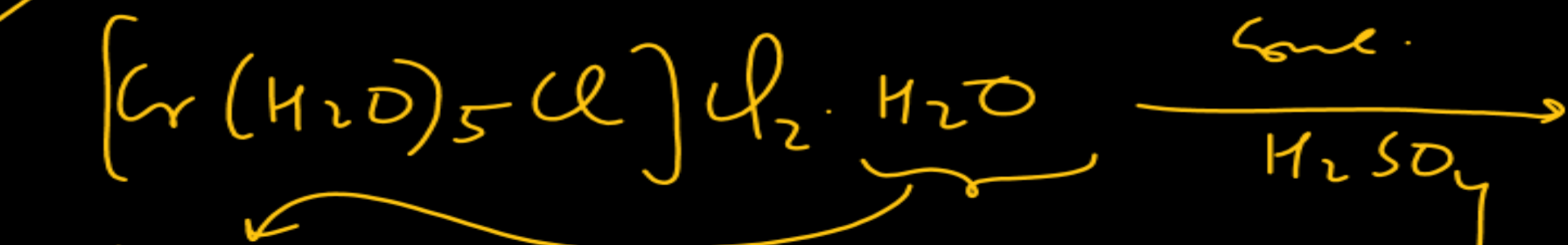
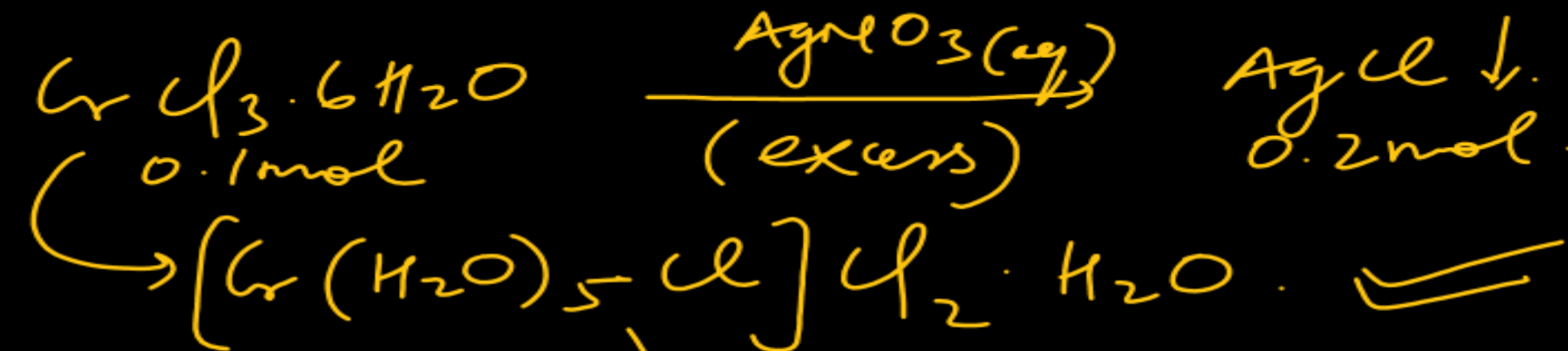
(At. Wt. Cr = 52, O = 16, Cl = 35.5, H = 1).

X (A) It will exist in 2 geometrical isomeric form

✓ (B) When solution of 0.1 mole complex passed in cationic exchanger resulting solution required 0.2 mole KOH for neutralisation

X (C) Diamagnetic complex in which central metal is in $d^2 sp^3$

✓ (D) It loses almost 6.75% of its original mass on treatment with conc. H_2SO_4 .



$\therefore \text{H}_2\text{O} = \frac{1 \times 18}{52 + 3 \times 35.5 + 6 \times 18} \times 100 \approx 6.75\%$

